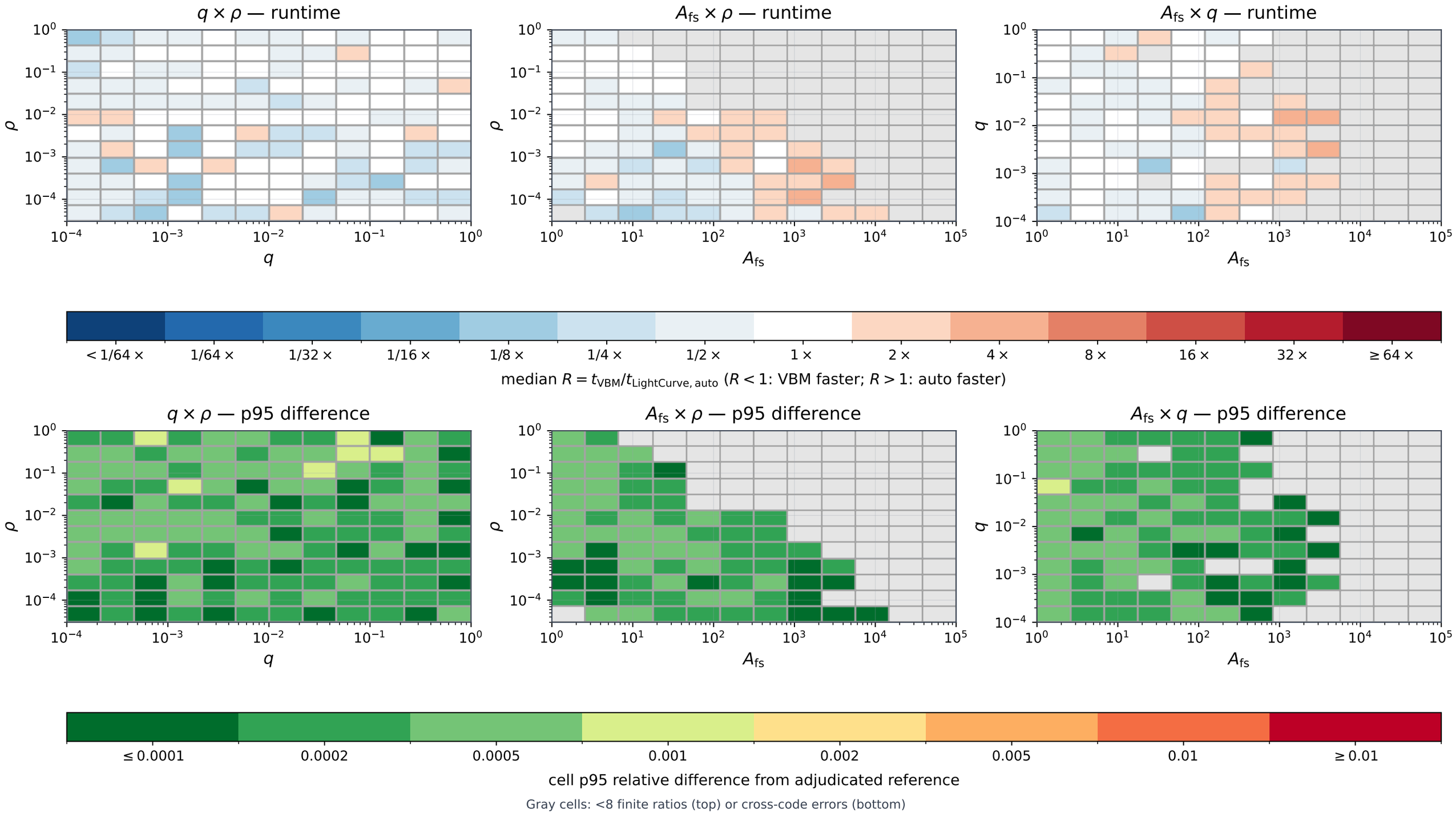linear limb darkening, $\varepsilon_{\mathrm{rel}} = 10^{-3}$